

# **PROOF / PROGRAM — Canonical Plate Folio**

## ONE CONSTRUCTIVE OBJECT, TWO PRESENTATIONS

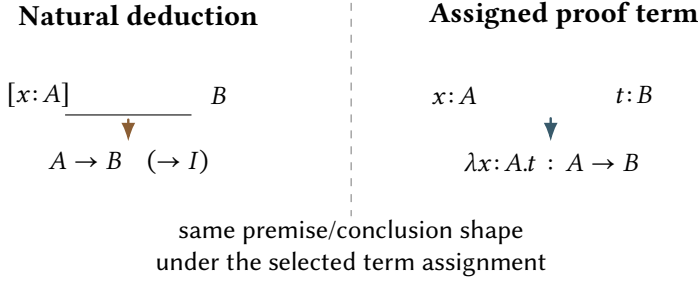


Figure 1: **Plate 1. One constructive object, two presentations.** Implication introduction discharges an assumption on the proof side and binds a variable on the term side. The alignment is exact for the selected CH-0 term assignment; it is not a claim that every proof formalism has this syntax.

## INTRODUCTION THEN ELIMINATION IS A DETOUR

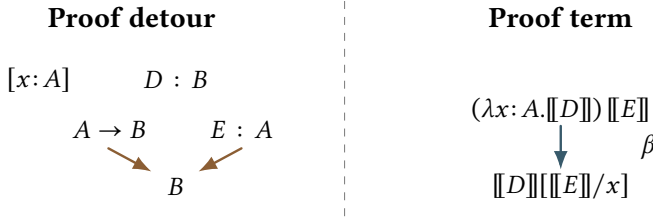


Figure 2: **Plate 2. Introduction then elimination is a detour.** The principal implication detour is assigned a beta redex, and proof substitution becomes capture-avoiding term substitution. This plate shows the principal implication case only; it does not prove global normalization.

## THE IMPLICATION MACHINE

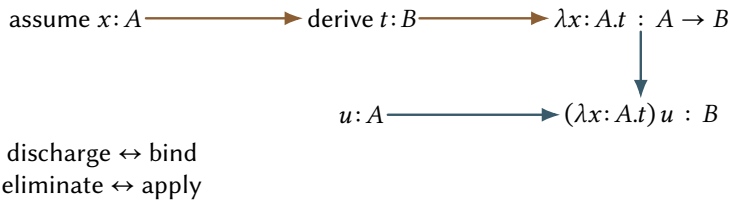


Figure 3: **Plate 3. The implication machine.** Assumption discharge becomes lexical binding, and modus ponens becomes application. The runtime reading does not replace semantic truth conditions; it exposes the constructive term carried by this calculus.

## A NORMALIZATION TRACE

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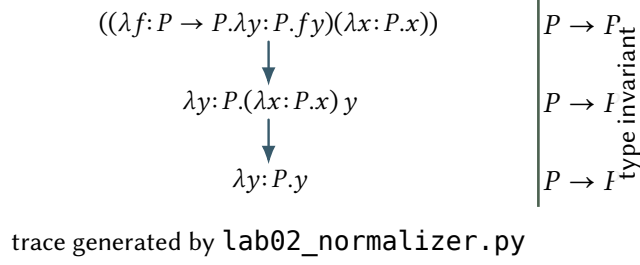
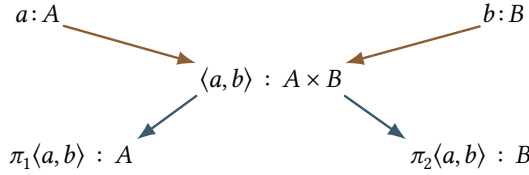


Figure 4: **Plate 4. A normalization trace.** The teaching normalizer contracts two beta redexes and rechecks the invariant type after each step. This computed trace is evidence for the implementation on the retained fixture; one trace is not a proof of strong normalization.

## CONJUNCTION / PRODUCT PREVIEW

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conjunction introduction carries both pieces of evidence

Figure 5: **Plate 5. Conjunction / product preview.** A constructive proof of conjunction carries evidence for both components, matching pair construction and projection in CH-0. The alignment is structural in this calculus, not a universal semantic identity between products and truth-valued conjunction.

## DISJUNCTION CARRIES A CHOICE

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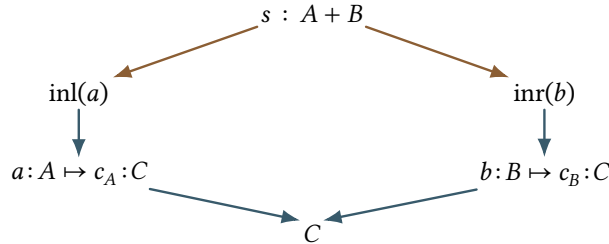


Figure 6: **Plate 6. Disjunction carries a choice.** Constructive disjunction records which injection was built and carries the evidence needed by the selected branch. This is richer than a proof-irrelevant Boolean truth value.

## PRODUCT DETOURS

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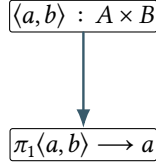


Figure 7: **Plate 7. Product detours.** Pair construction immediately followed by projection contracts to the selected component. Scope: principal CH-0 product reduction only.

## PRODUCT CANONICAL FORM

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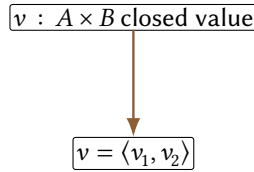


Figure 8: **Plate 8. Product canonical form.** A closed value of product type must be a pair under the CH-0 value grammar. Scope: a typing inversion fact, not a claim about arbitrary runtime objects.

## BOTH PIECES SURVIVE

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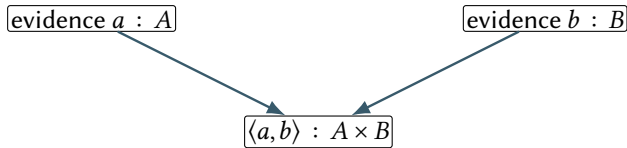


Figure 9: **Plate 9. Both pieces survive.** Constructive conjunction packages both witnesses, making projection computationally meaningful. Scope: no global proof-irrelevance conclusion is drawn.

## DISJUNCTION MAKES A CHOICE

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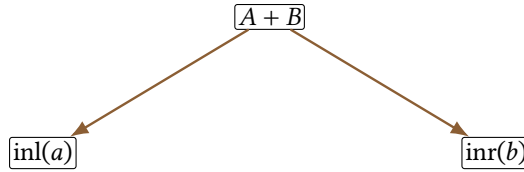


Figure 10: **Plate 10. Disjunction makes a choice.** A constructive sum inhabitant records a branch and its payload. Scope: richer than a proof-irrelevant Boolean, but not a universal semantics of disjunction.

## CASE BRANCHES CONVERGE

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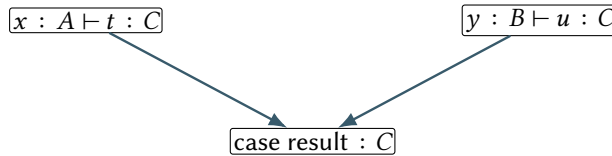


Figure 11: **Plate 11. Case branches converge.** Sum elimination may inspect either constructor, but both branches must produce the same result type. Scope: the stated CH-0 rule.

## SUM DETOUR

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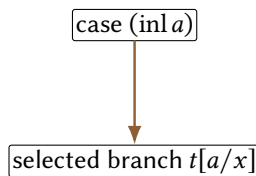


Figure 12: **Plate 12. Sum detour.** Injection followed immediately by case analysis contracts to the selected branch by substitution. Scope: principal reduction only; no arbitrary-extension confluence claim.

## TRUTH HAS A CONSTRUCTOR

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$\star : \mathbf{1}$

Figure 13: **Plate 13. Truth has a constructor.** The unit type has the canonical value  $\star$ , giving the constructive reading of truth. Scope: this does not imply equality of arbitrary proof objects at other types.

## EMPTY HAS NO CONSTRUCTOR

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no closed value  $v : \mathbf{0}$

↓

$e : \mathbf{0} \Rightarrow \text{abort}_A(e) : A$

Figure 14: **Plate 14. Empty has no constructor.** Empty elimination consumes an impossible premise; it does not fabricate a closed inhabitant of  $\mathbf{0}$ . Scope: the CH-0 canonical-form argument.

## CANONICAL FORMS BY TYPE

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|                            |
|----------------------------|
| $A \rightarrow B$ : lambda |
| $A \times B$ : pair        |
| $A + B$ : injection        |
| $\mathbf{1}$ : $\star$     |
| $\mathbf{0}$ : none        |

Figure 15: **Plate 15. Canonical forms by type.** Outer type constrains the possible shape of a closed value and powers the progress proof. Scope: scoped to the declared CH-0 value grammar.

## UNIVERSAL CONSTRUCTION

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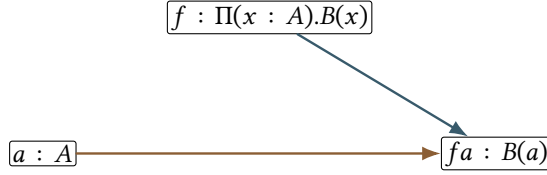


Figure 16: **Plate 16. Universal construction as dependent function.** A  $\Pi$ -term supplies evidence at each argument. Scope: the inherited CHD-1 rules, not every semantics of universal quantification.

## EXISTENTIAL EVIDENCE

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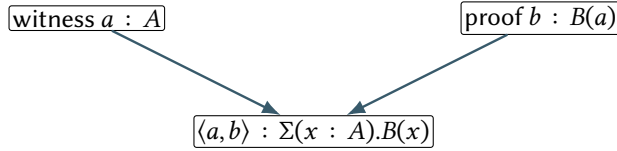


Figure 17: **Plate 17. Existential evidence is witness plus proof.** A dependent sum carries both a selected witness and evidence specialized to it. Scope: witness erasure requires a separate policy.

## DEPENDENT SUBSTITUTION

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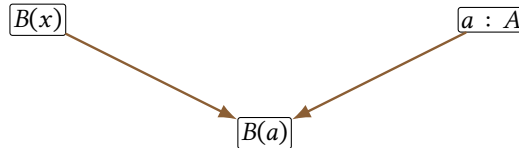


Figure 18: **Plate 18. Quantifier substitution.** Instantiation substitutes a witness into the dependent proposition/type family. Scope: capture avoidance and well-formed dependency remain formal obligations.

## TWO PROOF PRESENTATIONS

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Figure 19: **Plate 19. Natural-deduction and sequent views.** Translations connect two proof presentations with different administrative structure. Scope: correspondence, not literal identity of derivations.

## CUT AS COMPOSITION

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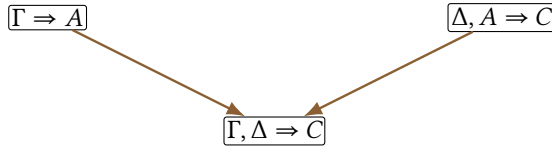


Figure 20: **Plate 20. Cut as explicit composition.** The cut formula is an explicit intermediate result consumed by the second derivation. Scope: LJ-0 only.

## CUT REDUCTION

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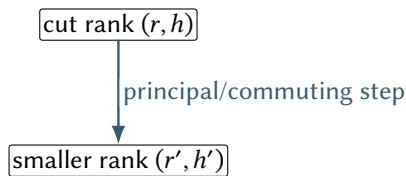


Figure 21: **Plate 21. Cut reduction trace.** A bounded cut step preserves the end sequent while decreasing the declared measure. Scope: computed trace is not the general cut-elimination theorem.



## A WORLD WHERE EXCLUDED MIDDLE FAILS

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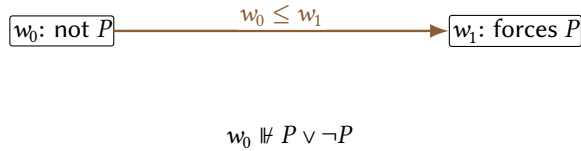


Figure 22: **Plate 22. Finite Kripke countermodel.** At the root, neither  $P$  nor  $\neg P$  is forced, so excluded middle fails there. Scope: selected intuitionistic Kripke semantics.

## CONSTRUCTIVE INFORMATION

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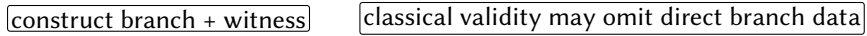


Figure 23: **Plate 23. Constructive information versus classical commitment.** Constructive introduction rules expose branch/witness data that classical validity need not provide in the same form. Scope: calculus-relative, not a value judgment.

## DOUBLE NEGATION IS A BOUNDARY

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Figure 24: **Plate 24. Double negation as a translation boundary.** Classical reasoning can enter a constructive target through a negative translation rather than literal proof identity. Scope: selected translation only.

## CONTINUATION-PASSING CONTROL

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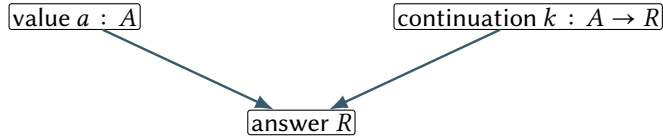


Figure 25: **Plate 25. Continuation-passing control flow.** CPS makes the future use of a value an explicit typed argument. Scope: the restricted CHC-1 translation, not pure CH-0.

## CLASSICAL PROOF AS CONTROL

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Figure 26: **Plate 26. Classical proof as control.** Selected classical principles receive computational readings through continuation structure. Scope: one typed interpretation among several, not universal identity.

## DIRECT STYLE / CPS

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Figure 27: **Plate 27. Direct style versus CPS.** A direct result becomes a computation parameterized by its continuation. Scope: typing preservation is proved only for the displayed restricted translation.

## RELEVANCE

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 $\text{inl}(a)$ 
 $\text{inr}(b)$ 

same sum type, potentially different behavior

Figure 28: **Plate 28. Relevance: two inhabitants, two behaviors.** Same-typed evidence can drive different branches in a relevance-sensitive context. Scope: no global proof-irrelevance assumption.

## ERASURE

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proof-bearing term  $\xrightarrow{\text{checked erasure}}$  retained program

Figure 29: **Plate 29. What erasure may remove.** Only structure proved irrelevant under the stated policy may disappear while retained behavior is preserved. Scope: toy marked fragment only.

## PROOF IRRELEVANCE IS ADDITIONAL

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 $\text{propositions-as-types}$ 
 $\text{does not imply all inhabitants equal}$ 

Figure 30: **Plate 30. Proof irrelevance is an additional principle.** Curry–Howard alone does not quotient inhabitants. Scope: systems may separately validate irrelevance in selected proposition universes.

## EXTRACTION PIPELINE

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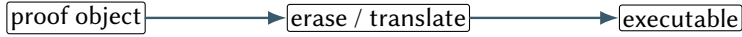


Figure 31: **Plate 31. Extraction pipeline.** A checked proof-bearing artifact is transformed into retained executable content. Scope: exact only for the declared extraction examples.

## WITNESS SURVIVES

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Figure 32: **Plate 32. Witness-preserving extraction.** Selected existential constructions retain the witness while proof certificates are checked and discarded. Scope: finite registered predicates only.

## EXTRACTION PRESSURE POINTS

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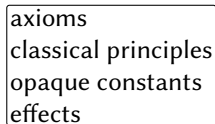


Figure 33: **Plate 33. Extraction pressure points.** Several ingredients require system-specific treatment before executable meaning is claimed. Scope: illustrative taxonomy, not a universal failure theorem.

## PARAMETRICITY AS UNIFORMITY

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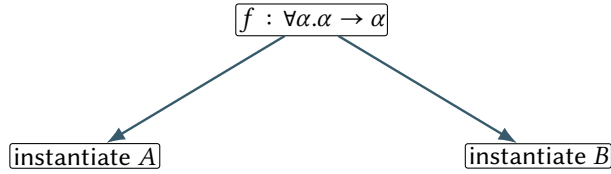


Figure 34: **Plate 34. Parametricity as uniformity.** A polymorphic term must act coherently across type instantiations under relational-parametricity hypotheses. Scope: CHF-1 preview.

## RELATIONS TRAVEL THROUGH TERMS

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Figure 35: **Plate 35. Relations travel through a polymorphic term.** Relational interpretation constrains paired outputs whenever paired inputs are related. Scope: imported theorem; finite lab checks only selected instances.

## FREE-THEOREM CONSTRAINT

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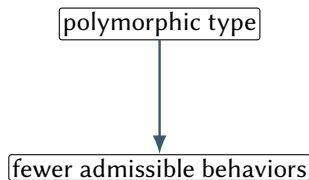


Figure 36: **Plate 36. Free-theorem constraint surface.** Parametric type structure can rule out implementations unavailable from ordinary monomorphic typing alone. Scope: purity and parametricity hypotheses are essential.

## TACTIC / ELABORATOR / KERNEL

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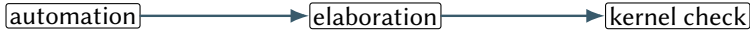


Figure 37: **Plate 37. Tactic, elaborator, kernel.** Convenience layers propose explicit objects; the kernel independently checks the final certificate. Scope: toy architecture abstracting real systems.

## THE TRUST CONE

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Figure 38: **Plate 38. The kernel trust cone.** Independent rechecking narrows authority to the checker, parser, calculus, runtime, and statement formalization. Scope: trust is concentrated, not eliminated.

## HOSTILE CERTIFICATE REJECTION

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Figure 39: **Plate 39. Hostile certificate rejection.** A malformed elaborator-shaped object fails independent kernel rechecking. Scope: demonstrates the laboratory boundary only.

## CLOSED TERM / OPEN CONVERSATION

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Figure 40: **Plate 40. Closed term versus open conversation.** Termination is natural for closed evaluation but can be the wrong success criterion for an interactive service. Scope: state-machine preview, not session typing.

## THE MISSING DUAL

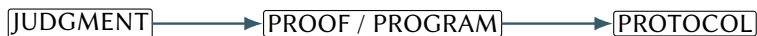
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Figure 41: **Plate 41. The missing dual.** Open interaction needs complementary endpoint obligations rather than one closed output type. Scope: previews session-type duality without formalizing it.

## THRESHOLD TO PROTOCOL

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channel · duality · protocol state · progress · deadlock

Figure 42: **Plate 42. Threshold to PROTOCOL.** The series moves from typed construction to typed interaction because closed normalization cannot express bilateral protocol obligations. Scope: the ten-volume unification remains a research thesis.